# Logarithms and Their Properties

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.02

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

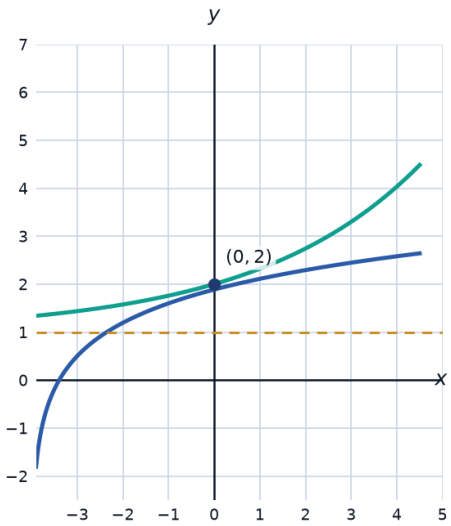
**1** **[Foundational]** • **Target: Evaluating logarithms**

Evaluate each logarithm exactly:

**a** $\log_2 32$ **b** $\log_3 \frac{1}{9}$ **c** $\log 1000$ **d** $\ln e^7$

**2** **[Foundational]** • **Target: Evaluating logarithms**

Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.




**3** **[Foundational]** • **Target: Evaluating logarithms**

State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

**4** **[Foundational]** • **Target: Properties of logarithms**

Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.

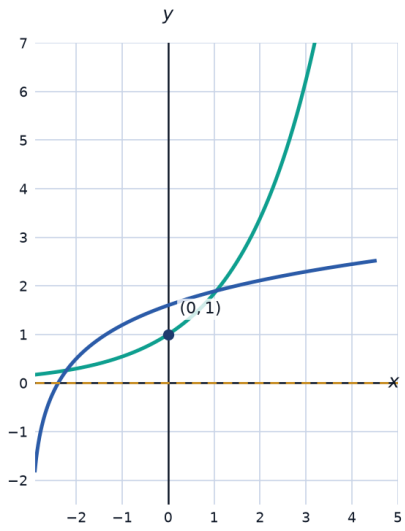
**5** **[Foundational]** • **Target: Properties of logarithms**

Write as a single logarithm: $2\log x + 3\log y - \log z$.

© ClipSAT — clipsat.org | All Rights Reserved

**6** [Foundational] • Target: Properties of logarithms

Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:




**a** $\log_b 6$ **b** $\log_b 8$ **c** $\log_b 1.5$

**7** [Foundational] • Target: Graphs of logarithmic functions

For a curriculum assessment problem, suppose for $y = \log_2(x + 4) - 1$, state:

**a** the domain **b** the vertical asymptote **c** the $x$-intercept

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Graphs of logarithmic functions

Describe how the graph of $g(x) = -\ln(x - 2) + 3$ is obtained from $y = \ln x$.

**9** [Intermediate] • Target: Logarithmic scales and applications

The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where $I$ is intensity and $I_0$ is a reference intensity.

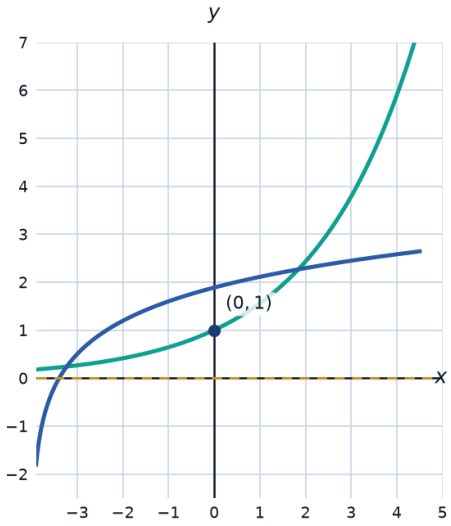
**a** An earthquake has $M = 6$. Express $I$ in terms of $I_0$.

**b** How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake?

© ClipSAT — clipsat.org | All Rights Reserved

**10** [Intermediate] • Target: Logarithmic scales and applications

Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does $\beta$ increase, to two decimal places?




**11** [Intermediate] • Target: More practice with logarithm properties

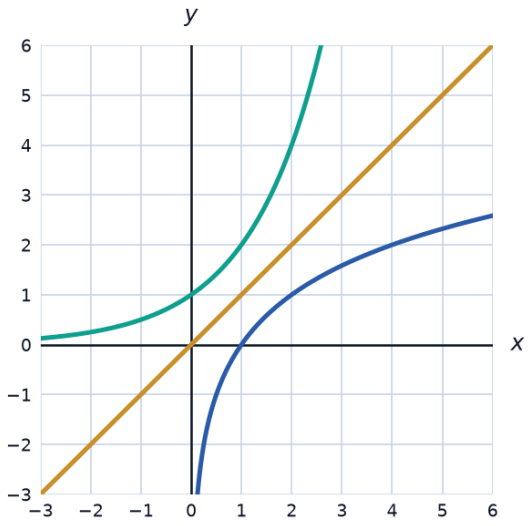
Expand fully: $\log_2\left(\frac{8x^2}{\sqrt[3]{y}}\right)$.

**12** [Intermediate] • Target: More practice with logarithm properties

Write as a single logarithm and simplify: $\frac{1}{2}\ln 16 + \ln 3 - \ln 2$.

**13** [Intermediate] • Target: Logarithms as inverse functions

The graphs of $y = 2^x$ and $y = \log_2 x$ are shown together with the line $y = x$. Explain why the two curves are reflections of each other across $y = x$.




**14** [Intermediate] • Target: Solving simple logarithmic equations

Solve $2\log x = \log(x + 6) + \log 1$ for $x$ (noting $\log 1 = 0$), and check for extraneous solutions.

© ClipSAT — clipsat.org | All Rights Reserved

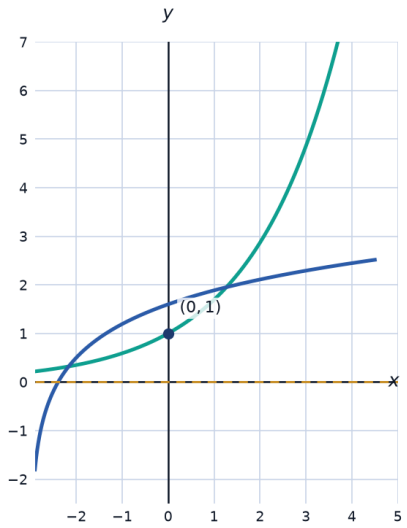
## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: The pH scale (a logarithmic application)** • **Cross-Lesson: 2.04 Law of Sines and Law of Cosines**

pH is defined as $pH = -\log[H^+]$, where $[H^+]$ is hydrogen ion concentration (mol/L). Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$, to two decimal places.

**16** **[Challenging]** • **Target: The pH scale (a logarithmic application)** • **Cross-Lesson: 3.01 The Unit Circle and Exact Values**

Two solutions have pH values of $3$ and $6$. Find the ratio of their hydrogen-ion concentrations $[H^+]$, and state which is more acidic.




**17** **[Challenging]** • **Target: Common mistakes with logarithm properties** • **Cross-Lesson: 3.02 Fundamental Trigonometric Identities**

Explain why $\log(x + y) \neq \log x + \log y$ in general, using a specific numerical counterexample.

**18** **[Challenging]** • **Target: Common mistakes with logarithm properties** • **Cross-Lesson: 3.03 Sum and Difference Formulas**

Simplify $\log_2(4x) - \log_2(2x)$ using the quotient property, and verify with a specific value of $x$.

**19** **[Challenging]** • **Target: Evaluating logarithms** • **Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations**

[Functions & Their Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

**20** **[Challenging]** • **Target: Properties of logarithms** • **Cross-Lesson: 4.01 Circles and Parabolas**

[Unit 2 · Exponential and Logarithmic Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

© ClipSAT — clipsat.org | All Rights Reserved

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Evaluating logarithms

Consider the functions $f(x) = \frac{1}{x-2}$ and $g(x) = 3x + k$. If $f(g(x)) = \frac{1}{2x-1}$, find the value of $k$.

**a.** 3

**b.** 2

**c.** 1

**d.** -1

**22** [Foundational] • Target: Properties of logarithms

Let $f(x) = \log_2(ax + b)$ and $g(x) = 3^x - 1$. Given that $f(g(x)) = x$ and the point $(2, 1)$ lies on the graph of $y = g(x)$, find the value of $a + b$.

**a.** 4

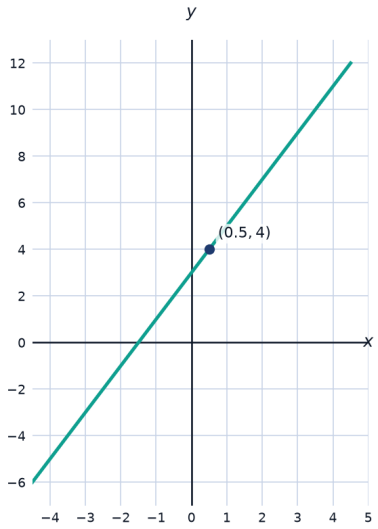
**b.** 3

**c.** 2

**d.** 1

**23** [Foundational] • Target: Graphs of logarithmic functions

Given the function $f(x) = 2x + 3$. What is the value of $f^{-1}(7)$?




**a.** 1

**b.** 2

**c.** 3

**d.** 4

© ClipSAT — clipsat.org | All Rights Reserved

**24** [Foundational] • Target: Logarithmic scales and applications

Consider the functions $f(x) = 3x + 2$ and $g(x) = x - 5$. What is the value of $f(g(3))$?

**a.** 1

**b.** -4

**c.** 13

**d.** -10

**25** [Foundational] • Target: More practice with logarithm properties

Which is the inverse of $f(x) = \log_2 x$?

**a.** $g(x) = \log_2 x$

**b.** $g(x) = 2x$

**c.** $g(x) = x^2$

**d.** $g(x) = 2^x$

**26** [Foundational] • Target: Logarithms as inverse functions

If $f(x) = x^2 + 1$ and $g(x) = -x - 1$, find the sum of all solutions to $(f \circ g)(x) = 5$.

**a.** -2

**b.** 4

**c.** -6

**d.** -10

**27** [Foundational] • Target: Solving simple logarithmic equations

Find the number of real solutions to $f(x) = x$ where $f(x) = x^3 - 3x$.

**a.** 1

**b.** 0

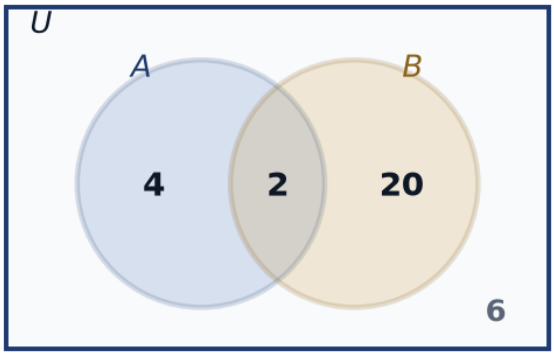
**c.** 3

**d.** 2

© ClipSAT — clipsat.org | All Rights Reserved

**28** [Foundational] • Target: The pH scale (a logarithmic application)

Find the number of surjections from a 4-element set to a 2-element set.




**a.** 14

**b.** 8

**c.** 16

**d.** 12

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Common mistakes with logarithm properties

Find all functions $f$ with $f(x)^2 = x^2$ for all $x$.

**a.** $f(x) = x$, $f(x) = -x$, or $f(x) = |x|$ or $f(x) = -|x|$

**b.** $f(x) = x$ only

**c.** $f(x) = 0$

**d.** $f(x) = \pm 1$

**30** [Intermediate] • Target: Evaluating logarithms

Find all polynomials $p(x)$ of degree 2 satisfying $p(x+1) - p(x) = 2x + 1$ and $p(0) = 0$.

**a.** $p(x) = 2x$

**b.** $p(x) = x^2 - x$

**c.** $p(x) = x^2 + x$

**d.** $p(x) = x^2$

© ClipSAT — clipsat.org | All Rights Reserved

**31** [Intermediate] • Target: Properties of logarithms

Solve: $3^x = 2^{x+1}$. Use logarithms.

**a.** $x = \frac{\ln 2}{\ln 3}$

**b.** $x = 1$

**c.** $x = \frac{\ln 3}{\ln 2}$

**d.** $x = \frac{\ln 2}{\ln 3 - \ln 2}$

**32** [Intermediate] • Target: Graphs of logarithmic functions

What value must be excluded from the domain of $f(x) = \frac{1}{3x+3}$?

**a.** 3

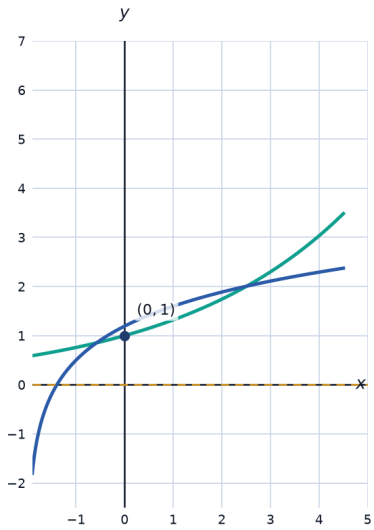
**b.** -1

**c.** -4

**d.** 2

**33** [Intermediate] • Target: Logarithmic scales and applications

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:




**a.** No conclusion possible

**b.** There exists $c \in (1, 4)$ with $f(c) = 0$

**c.** $f(2) = 1$

**d.** $f$ is constant

© ClipSAT — clipsat.org | All Rights Reserved

**34** **[Intermediate]** • **Target: More practice with logarithm properties**

If $f(x) = -2x + 1$ and $g$ is the inverse of $f$, find $g(f(-1))$.

**a.** 4

**b.** -1

**c.** 2

**d.** -10

**35** **[Intermediate]** • **Target: Logarithms as inverse functions**

Find a value of $c$ making $f$ continuous at $x = 2$, where $f(x) = \{cx + 1, x < 2; x^2 - 1, x \geq 2$.

**a.** $c = 1$

**b.** $c = 3$

**c.** $c = 0$

**d.** $c = 2$

**36** **[Intermediate]** • **Target: Solving simple logarithmic equations**

The Banach fixed-point theorem guarantees a unique fixed point when $f$ is:

**a.** Any continuous function

**b.** Lipschitz with $L = 1$

**c.** A bijection

**d.** A contraction mapping on a complete metric space

**37** **[Intermediate]** • **Target: The pH scale (a logarithmic application)**

Solve: $2^{x+1} = 5$. Express $x$ using logarithms.

**a.** $x = \frac{\ln 5}{\ln 2} - 1$

**b.** $x = \frac{\ln 5 - 1}{\ln 2}$

**c.** $x = \log_2 5 - 1$

**d.** Both A and C

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

**38** [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.03 Sum and Difference Formulas

Find all rational functions $f(x) = \frac{ax+b}{cx+d}$ that are their own inverse.

**a.** $b = c$

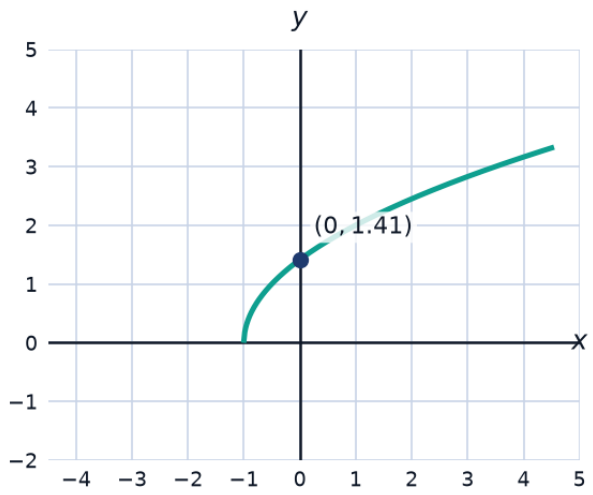
**b.** $f(f(x)) = x$ impossible for rational

**c.** $a = d$

**d.** $a = -d$ (i.e. $a + d = 0$)

**39** [Challenging] • Target: Evaluating logarithms • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

What is the smallest integer in the domain of $f(x) = \sqrt{2x+2}$?




**a.** -8

**b.** -9

**c.** -1

**d.** -10

**40** [Challenging] • Target: Properties of logarithms • Cross-Lesson: 4.01 Circles and Parabolas

If $f(x) = \sqrt{9 - x^2}$, what is the shape of its graph?

**a.** Parabola

**b.** Upper semicircle of radius 3

**c.** Full circle

**d.** Hyperbola

© ClipSAT — clipsat.org | All Rights Reserved

**41** [Challenging] • Target: Graphs of logarithmic functions • Cross-Lesson: 4.02 Ellipses

Find the interval where $f(x) = x^2 - 4x + 3$ is positive.

**a.** $x > 0$

**b.** $1 < x < 3$

**c.** $x < 0$

**d.** $x3$

**42** [Challenging] • Target: Logarithmic scales and applications • Cross-Lesson: 4.03 Hyperbolas

The Weierstrass function is continuous everywhere and:

**a.** Differentiable everywhere

**b.** Differentiable at integers only

**c.** Differentiable nowhere

**d.** Piecewise linear

**43** [Challenging] • Target: More practice with logarithm properties • Cross-Lesson: 5.01 Vectors: Components and Operations

Which describes a vertical stretch of $y = f(x)$ by factor 4?

**a.** $y = f(4x)$

**b.** $y = f(x/4)$

**c.** $y = f(x) + 4$

**d.** $y = 4f(x)$

**44** [Challenging] • Target: Logarithms as inverse functions • Cross-Lesson: 5.02 The Dot Product

Which function is the inverse of $f(x) = e^x$?

**a.** $g(x) = x^e$

**b.** $g(x) = \ln x$

**c.** $g(x) = 10^x$

**d.** $g(x) = \log x$

**45** [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 5.03 Parametric Equations

A function has zeros at $x = -1, 0, 2$ and positive leading coefficient. Sketch shows which end behavior?

**a.** Left: $f \to +\infty$; Right: $f \to -\infty$

**b.** Left: $f \to -\infty$; Right: $f \to +\infty$

**c.** Both ends to $+\infty$

**d.** Both ends to $-\infty$

© ClipSAT — clipsat.org | All Rights Reserved